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Effectiveness of health literacy education on health literacy in early adolescence: A randomized controlled trial

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ABSTRACT

Objectives: The present study was conducted to measure the effectiveness of health literacy education on the health literacy level of children in early adolescence.

Study design: A parallel group randomized controlled trial was conducted.

Methods: The study group consisted of sixth and seventh-grade students (aged 12–13 years) studying in a secondary school in Ankara. The study was completed by 80 students, 40 in the intervention group and 40 in the control group. The Health Literacy Education program, consisting of four sessions, was presented to the intervention group. The instruments were administered to both groups before the intervention (pre-test), one month (post-test), and three months after the intervention (follow-up test). The data was analyzed using the chi-square test, and the two-way mixed-design variance analysis with Bonferroni Multiple Comparison Test.

Results: Before the study, the adolescents in the intervention and control groups were similar regarding descriptive characteristics ($p > 0.05$). There was no statistically significant difference between the pre-test health literacy mean scores of the intervention and control groups ($p > 0.05$). After the health literacy education, health literacy post-test scores and follow-up test mean scores of the intervention group increased significantly compared to the control group ($p < 0.05$). The intervention had a large effect size ($F = 414.683$; $\eta^2 = 0.91$) at the 95 % confidence interval.

Conclusions: The study revealed that health literacy education was effective in increasing the level of health literacy in early adolescence. It is recommended that health literacy education be added to the school education curriculum of adolescents.

Introduction

Health literacy (HL) is critical to give adolescents the responsibility to increase their self-efficacy and improve their health.¹ Recent studies examining the level of HL in adolescents revealed that the HL level of adolescents is insufficient.^{2–4} A study conducted in European countries determined that 80.5 % of adolescents had low and medium health literacy.³ An Italian study found that the level of health literacy needed to be improved or more problematic in 93.2 % of adolescents.⁴ A study conducted in Türkiye found that 58.8 % of adolescents had inadequate or problematic HL.³ Therefore, effective interventions to increase the health literacy level of adolescents need to be implemented.

Health education is a significant opportunity to improve health literacy, make the right decisions about health, and identify the concept of health and its affecting factors.⁵ Health education is manageable,

cost-effective, and provides the opportunity to reach many people.⁶ Due to the proven positive effects of health education on improving health literacy in late adolescence,⁷ it is predicted that it can be a practical approach to increase health literacy in early adolescence.

Since behavioral development has just begun in adolescence, it is easier to make interventions to improve the level of HL compared to adulthood.⁸ There are a limited number of intervention studies aimed to increase health literacy in adolescents.^{6,9} Studies show that although educational interventions positively contribute to increasing health literacy, there is still a need for interventions to increase the level of health literacy in early adolescents. It is estimated that this study would provide strong evidence to demonstrate the positive effect of health education given to children in early adolescence on health literacy and guide other health professionals working in the field, especially school nurses.

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The current study was conducted to measure the effectiveness of health literacy education in improving the health literacy of children in early adolescence. The research hypothesis that "There is no difference between the intervention group given health literacy education and the control group regarding their health literacy level."

Methods

Study design

The study was conducted with a parallel group randomized controlled trial and reported based on the CONSORT (Consolidated Standards of Reporting Trials) Non-Pharmacological Treatment checklist.

Participants

The study group consisted of sixth and seventh-grade students (aged 12–13 years) studying at a secondary school in a city in Türkiye (N = 727). A power analysis was performed using the health literacy variable from a previous study to determine the sample size.⁷ Accordingly, the sample size was calculated as at least 73, based on $\alpha = 0.05$, $1-\beta = 0.95$, and $d = 0.43$. Due to possible dropouts, the sample was increased by 10 %, and 80 adolescents who met the inclusion criteria were included in the study.

To evaluate the interaction of the intervention and control groups in the study, only the control II group was created from a different secondary school with similar sociocultural characteristics and with institutional permission. Forty adolescents who met the inclusion criteria were included in the study. At the end of the study, the data of control group I and control group II were compared, and all measurements were similar. Therefore, it was found that there was no interaction between the intervention and control groups.

Inclusion criteria were: (1) being in the sixth or seventh grade, (2) having parental approval, (3) volunteering to participate in the study, and (4) having a low level of health literacy. Exclusion criteria for selection were: (1) difficulty in reading and writing Turkish, (2) having a medium or high level of health literacy, and (3) having a physical/mental disability or illness. Exclusion criteria from the study were: (1) failure to attend at least two training sessions and (2) failure to complete the instruments.

Randomization and prevention of bias

The adolescents were assigned to the intervention and control groups by the stratified randomization method. To ensure homogeneity between groups, adolescents were divided into four strata according to gender (female, male) and grade level (sixth and seventh). Adolescents from the created strata were divided into intervention and control groups in a 1:1 ratio. An independent statistician created a series of random numbers on the EXCEL program, and the adolescents were assigned to intervention (n = 40) and control (n = 40) groups with a simple randomization method.

To prevent selection bias in the study, adolescents were assigned to the intervention and control groups by randomization. The assignment of adolescents to groups was made by a statistician who was outside the research team. To prevent reporting bias, the data were evaluated by a statistician. The data were coded and transferred to the computer without specifying the intervention and control groups. Since one of the researchers conducted education sessions, blinding of the participants and researcher could not be done.

Instruments

A personal information form and the Health Literacy Scale for School-Age Children were used to collect data. The personal information

form was prepared by the researchers based on the literature.^{3,10} The form consisted of nine closed-ended questions to determine the students' academic level, age, gender, parent's education level, parent's employment status, family income level, and academic achievement.

The Health Literacy Scale for School-Age Children was developed by Paakkari et al., in 2016.⁸ Scale items have a four-point Likert-type response option (definitely not true = 1, not completely true = 2, somewhat true = 3, and absolutely true = 4). The total score obtained by summing the items varies between 10 and 40 points. The scale score defines low (10–25 points), medium (26–35 points), and high (36–40 points) health literacy.⁸ The Turkish validity and reliability study was conducted by Haney in 2018,¹¹ and the Cronbach Alpha reliability coefficient was found to be 0.77. In this study, the Cronbach Alpha reliability coefficient was found to be 0.92 for the post-test and 0.94 for the follow-up test.

Procedure and measurements

The study was conducted between September 2022 and January 2023. The school's administration and teachers were informed about the implementation process of the study. Informed consent forms for participation of the children were sent to the parents in a sealed envelope. Adolescents were asked to deliver the envelopes received from their parents to their classroom teachers within a week. The personal information form and the Health Literacy Scale for School-Age Children (pre-test) were administered to the students who received consent from their families and volunteered to participate in the study. After the evaluation, those in the low (n = 207), medium (n = 93), and high (n = 427) HL categories were determined according to the pre-test mean score of the Health Literacy Scale for School-Age Children. The scale was re-administered to the adolescents one month (post-test) and three months (follow-up test) after the education program. Instruments were administered under the observation of the researcher. The average response time to instruments was 15 min.

Intervention

A health literacy education program was presented to the intervention group. This program was prepared by researchers based on the literature^{5,7,13,14,16} to provide adolescents with knowledge and attitudes about health literacy. The content of the education program was prepared in accordance with the age and developmental characteristics of the adolescents, and no medical terms were used. After the pilot study was carried out with five adolescents, the education program was finalized.

Intervention group

The health literacy education program was presented in the school conference hall to the intervention group by one of the researchers. Four education sessions were administered to the intervention group, one week apart. Each session lasted an average of 40 min. Educational methods and techniques such as narration, question-answer, group discussion, whisper technique, six thinking hats technique, and case study were used in the education sessions. Tools such as computers, projectors, blackboards, pencils, paper, and colored papers were used.

Control group

No other intervention was given to the control group besides the health-related information in the standard education curriculum. The personal information form and the Health Literacy Scale for School-Age Children were given to the students in the control group in parallel with the intervention group. After the study was completed, health education was planned for the control group, however, no students requested it. The educational content was turned into a brochure and delivered to the

school administration to give any school adolescents who requested it.

Data analysis

The IBM SPSS (Statistical Package for Social Sciences, Armonk, NY; IBM Corp.) 25.0 package program was used to analyze the data. Health literacy level was the outcome variable, and the health education program was the independent variable. The normality distribution of the data was evaluated with the Shapiro-Wilk test and kurtosis and skewness coefficients of ± 1.5 , and it was determined that they were by the normal distribution. Number, percentage, mean, and standard deviation (SD) were used to present the descriptive characteristics of the adolescents. Categorical variables were compared with the chi-square test. The Two-Way Mixed-Design Variance Analysis with the Bonferroni Comparison Test was used to examine the group, time, and group*time interaction of the measurements of the intervention and control groups. The first factor was groups (intervention and control), and the second was time intervals (pre-test, post-test, and follow-up test). Partial eta squared (η^2) value was used to calculate the effect size. Eta squared is considered as ≥ 0.01 small effect, ≥ 0.06 medium effect, and ≥ 0.14 large effect (Cohen and Dacanay, 1992). The significance level was accepted as $p < 0.05$.

Ethical dimension

Ethics committee approval and institutional permission were obtained before the study. Informed assent and consents were obtained from the adolescents and their parents. The content of the education program given to the intervention group was turned into a brochure and delivered to the school administration. Thus, students who requested it could use it for informational purposes.

Results

Among the recruited adolescents, those who did not speak Turkish ($n = 26$), were not at school on the day of the pre-test ($n = 29$), had high ($n = 93$) and medium ($n = 427$) health literacy levels, were not given consent by their parents ($n = 71$), and those who declared that they did not want to participate ($n = 1$) were excluded from the study (Fig. 1).

There was no statistically significant difference between the intervention and control groups regarding gender, academic level, parental education level and employment status, income, and academic achievement ($p > 0.05$) (Table 1).

The two-way mixed-design variance analysis revealed that the difference between groups (intervention and control) and time intervals (pre-test, post-test, follow-up test) ($p < 0.001$) and the interaction of

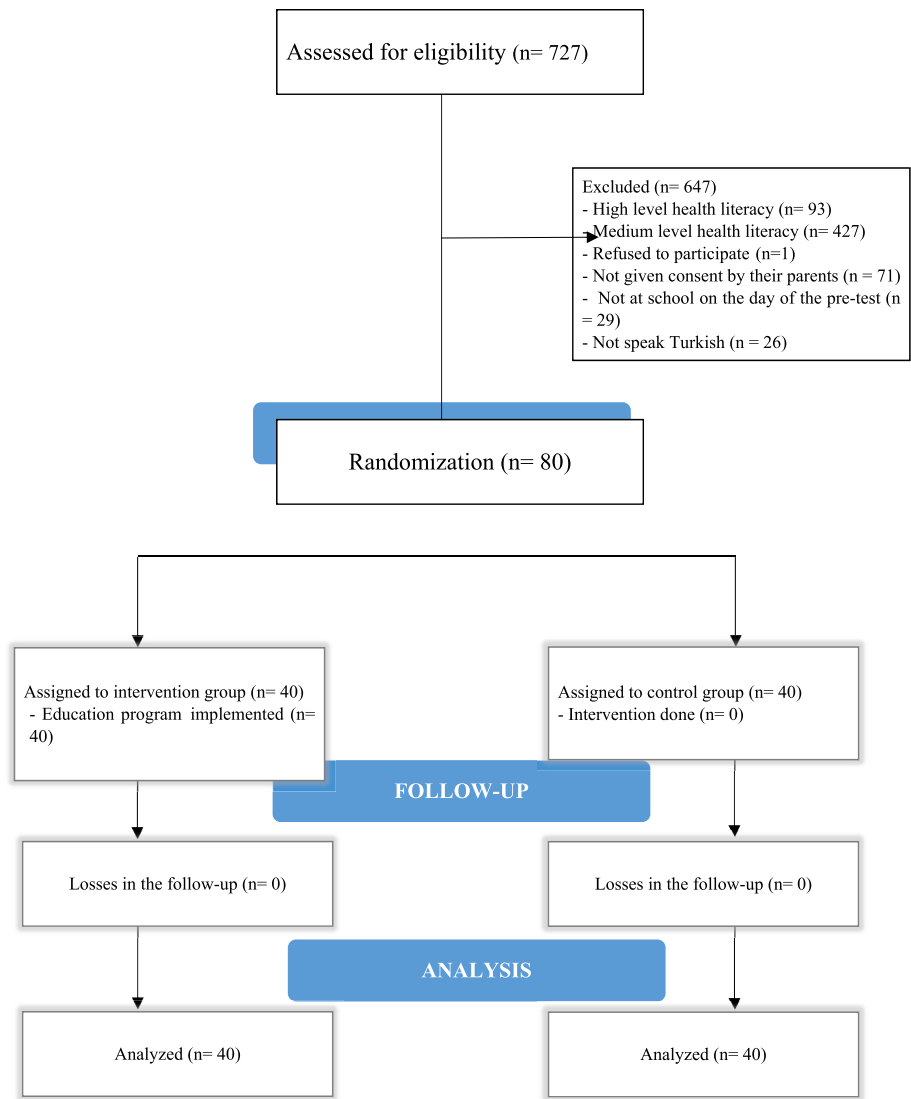


Fig. 1. CONSORT flow diagram.

Table 1
Individual characteristics of the intervention and control groups.

Individual characteristics	Intervention group (n = 40)		Control group (n = 40)		Statistical analysis
	n	%	n	%	
Gender					
Girl	20	50.0	20	50.0	$\chi^2 = 0.000$
Boy	20	50.0	20	50.0	$p = 1.000$
Academic level					
Sixth-grade	20	50.0	20	50.0	$\chi^2 = 0.558$
Seventh-grade	20	50.0	20	50.0	$p = 0.443$
Mother's education level					
Primary/secondary	17	47.5	19	45.0	$\chi^2 = 0.202$
High school/university	23	57.5	21	55.0	$p = 0.653$
Father's education level					
Primary/secondary	17	47.5	15	37.5	$\chi^2 = 0.208$
High school/university	23	57.5	25	62.5	$p = 0.648$
Mother's working status					
Working	15	37.5	13	32.5	$\chi^2 = 0.220$
Not working	25	62.5	27	67.5	$p = 0.639$
Father's working status					
Working	35	87.5	37	92.5	$\chi^2 = 0.556$
Not working	5	12.5	3	7.5	$p = 0.456$
Income					
Less than expenses	8	20.0	3	7.5	$\chi^2 = 4.196$
Equal to expenses	28	70.0	28	70.0	$p = 0.123$
More than expenses	4	10.0	9	22.5	
Academic achievement					
Good	11	27.5	16	40.0	$\chi^2 = 1.398$
Moderate	29	72.5	24	60.0	$p = 0.237$

χ^2 = Chi-squared test.

Table 2
Health literacy mean scores of the intervention and control groups and two-way mixed-design variance analysis results.

Measurements	Intervention group (n = 40)	Control group (n = 40)		Group	Time	Group*Time interaction
	Mean (SD)	Mean (SD)				
Health literacy						
Pre-test	21.75 (2.56) ^a	21.52 (3.05) ^a	F	1122.30	161.744	414.683
Post-test	34.00 (1.68) ^b	19.50 (2.16) ^a	P	<0.001	<0.001	<0.001
Follow-up test	34.23 (1.12) ^c	17.80 (1.44) ^a	η^2 (%95 CI)	0.95 (0.95–0.97)	0.81 (0.72–0.85)	0.91 (0.88–0.93)

Note: Different lowercase letters in each column indicate significant differences between measurement times ($p < 0.05$).
SD= Standard Deviation, CI= Confidence Interval.
F = Two-Way Mixed-Design ANOVA, η^2 = Partial Eta Squared.

group and time ($p < 0.001$) were significant for health literacy. The pre-test mean score of the intervention group was 21.75 (SD = 2.56), the post-test mean score (first month) was 34.00 (SD = 1.68), and the follow-up test mean score (third month) was 34.23 (SD = 1.12). The pre-test mean score of the control group was 21.52 (SD = 3.05), the post-test mean score (first month) was 19.50 (SD = 2.16), and the follow-up test

mean score (third month) was 17.80 (SD = 1.44). The mean health literacy scores after the education program in the intervention group increased significantly compared to the control group ($p < 0.05$). The intervention had a large effect size ($\eta^2 = 0.91$), and power level (99.9 %) within the 95 % confidence interval (Table 2) (Fig. 2).

Discussion

Health education is a crucial method to increase health literacy.⁷ The present study revealed that after the education program presented to improve health literacy in early adolescence, the health literacy level of the intervention group increased significantly compared to the control group. The activities such as explaining health literacy to students comprehensively, discussing the subject with sample scenarios, and using interactive methods in education sessions were effective in increasing the health literacy level of the students in the intervention group. Moreover, the health literacy level of the control group continued to decrease over time. This may be due to differences in abilities or opportunities to access and acquire information or challenges of using health information due to a lack of supportive environments and resources for implementing healthy habits.

There is limited evidence in the literature that education programs increase health literacy.^{6,9,12} In the study conducted by de Buhr et al.,⁹ after the basic care and health education given by school health nurses in schools for 9–12 months, the rate of adolescents with low and medium

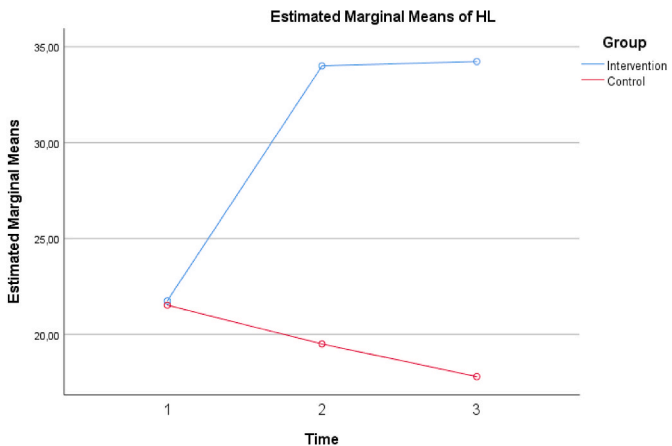


Fig. 2. Change over time in health literacy mean scores of intervention and control groups.

health literacy levels decreased from 17.9 % to 14.9 %. Dülger and Ayaz-Alkaya⁶ revealed that the health literacy levels of adolescents increased after the web-based education program grounded on health literacy. Akca and Ayaz-Alkaya¹² found that motivational interviewing and health education increased health literacy. Therefore, intervention studies to increase adolescent health literacy need to be implemented. In this context, it becomes essential for the nurses to protect and improve adolescent health actively.¹³ It is thought that nurses could contribute to the improvement of adolescents' health literacy levels by using their roles as educators, consultants, and case managers.

Health literacy studies have recently increased among adolescents due to the integration of health literacy strategies early in life.^{14,15} A scoping review of multiple studies concluded that adolescents and young adults have low to moderate levels of health literacy.¹⁵ Studies investigating the health literacy level of adolescents show that health literacy is mostly at low or medium levels.^{2,3,16,17} Although the importance of health literacy increases in adolescence, it appears that the health literacy of adolescents needs to be improved. Increasing the level of health literacy will help adolescents adopt a healthy lifestyle and, therefore, become healthy adults in the future.¹⁵ It is thought that initiatives to improve health literacy should be started as early as possible. School nurses, who have the opportunity to reach many people, are in a vital position with all the resources they possess, especially education, to improve adolescent health literacy.¹³ Nurses could significantly contribute to the development of adolescent health by teaching adolescents how to access accurate information and developing their ability to make correct health decisions at an early age.

Increased health literacy may increase adolescents' likelihood of adopting healthy lifestyles and having good physical and psychological health in adulthood.¹⁵ In this study, according to the follow-up test conducted in the third month after the education program, it was revealed that the health literacy level of the intervention group continued to increase, and the intervention had a large effect size. This finding is crucial as it shows that health literacy education can have positive effects after three months. In addition, this finding increased the reliability of the study and provided substantial evidence for other intervention studies involving health education in early adolescence in the future. It is known that health literacy is associated with health behaviors^{1,10,16,18} and self-efficacy.^{1,18} Therefore, it is estimated that health literacy education would contribute to increasing the health literacy level of adolescents and thus helping them acquire healthy lifestyle habits.

Strengths and limitations

This study contributes to the literature as one of the first evidence revealing the effect of the health literacy education program on the health literacy of children in early adolescence. Choosing a second control school to evaluate the interaction between the intervention and control group in the study is another strength. Conducting the study with a randomized controlled trial, applying a follow-up test in the third month after the intervention, and providing effect sizes and confidence intervals in reporting are other strengths of the study.

The study only includes data on sixth and seventh-grade students at these schools. Responses to data collection tools are limited to students' statements. Since the study implementation process was carried out outside of class, there was difficulty in ensuring the volunteer basis.

Conclusion

The results showed that the research hypothesis was rejected. Accordingly, the present study revealed that health literacy education is an effective method for increasing the health literacy levels of adolescents. Health education programs aimed at improving health literacy in adolescents should be disseminated. Different teaching methods (such as games, theater, and puzzles) could be prepared and used in education to

improve health literacy. Measurements could be performed regularly to determine the level of health literacy education programs. Experimental studies using different intervention methods (such as web-supported, simulation, and animation) to increase adolescent health literacy should be planned in the future.

Author statements

Ethical approval

The study was approved by the Gazi University Ethics Commission (08.03.2022–16). The study was registered to the [ClinicalTrials.gov](https://clinicaltrials.gov) database (NCT06231667).

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Competing interests

The authors report no actual or potential conflict of interest.

Data availability

The data for this study are available from the corresponding author upon reasonable request.

Patient consent

Informed assent and consents were obtained from the adolescents and their parents.

CRediT authorship contribution statement

Büşra Nur Yüksek: design, Data collection, Formal analysis, Manuscript writing, Writing – review & editing, for important intellectual content. **Sultan Ayaz-Alkaya:** design, Formal analysis, Supervision, Manuscript writing, Writing – review & editing, for important intellectual content.

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References

- Bektas İ, Kudubeş AA, Ayar D, Bektas M. Predicting the healthy lifestyle behaviors of Turkish adolescents based on their health literacy and self-efficacy levels. *J Pediatr Nurs*. 2021;59:20–25. <https://doi.org/10.1016/j.pedn.2021.01.016>.
- Haney MO. Health literacy and predictors of body weight in Turkish children. *J Pediatr Nurs*. 2020;55:257–262. <https://doi.org/10.1016/j.pedn.2020.05.012>.
- Paakkari L, Torppa M, Mazur J, et al. A comparative study on adolescents' health literacy in Europe: findings from the HBSC study. *Int J Environ Res Publ Health*. 2020;17(10):3543. <https://doi.org/10.3390/ijerph17103543>.
- Velasco V, Gragnano A, Gruppo Regionale HBSC Lombardia, Vecchio LP. Health literacy levels among Italian students: monitoring and promotion at school. *Int J Environ Res Publ Health*. 2021;18(19):9943. <https://doi.org/10.3390/ijerph18199943>, 2018.
- Paakkari L, Inchley J, Schulz A, Weber MW, Okan O. Addressing health literacy in schools in the WHO European region. *Public Health Panorama*. 2019;5(2-3).
- Dülger, H., & Ayaz-Alkaya, S. The effect of health literacy-grounded web-based education on nutrition and exercise behaviours in adolescents: A randomized controlled trial. *Int J Nurs Pract*, e13253. <https://doi.org/10.1111/ijn.13253>.
- Akca A, Ayaz-Alkaya S. Effectiveness of health literacy education for nursing students: a randomized controlled trial. *Int J Nurs Pract*. 2021;27(5), e12981. <https://doi.org/10.1111/ijn.12981>.
- Paakkari O, Torppa M, Kannas L, Paakkari L. Subjective health literacy: development of a brief instrument for school-aged children. *Scand J Publ Health*. 2016;44(8):751–757. <https://doi.org/10.1177/1403494816669639>.

9. de Buhr E, Ewers M, Tannen A. Potentials of school nursing for strengthening the health literacy of children, parents and teachers. *Int J Environ Res Publ Health*. 2020; 17(7):2577. <https://doi.org/10.3390/ijerph17072577>.
10. Ozturk FO, Ayaz-Alkaya S. Health literacy and health promotion behaviors of adolescents in Turkey. *J Pediatr Nurs*. 2020;54:31–35. <https://doi.org/10.1016/j.pedn.2020.04.019>.
11. Haney MO. Psychometric testing of the Turkish version of the health literacy for school-aged children scale. *J Child Health Care*. 2018;22(1):97–107. <https://doi.org/10.1177/1367493517738124>.
12. Akca A, Ayaz-Alkaya S. The effect of health literacy-based motivational interviewing and health education on adolescents' nutrition and exercise behaviors. In: *A randomized controlled study*. Ankara: Presented conference paper in 8th International 19th National Nursing Congress; 2024:25–28. September.
13. Akca A, Ayaz-Alkaya S. Health Literacy in the school-age period and the role of the school health nurse. *Journal of Hacettepe University Faculty of Nursing*. 2021;8(3): 328–334. <https://doi.org/10.31125/hunhemsire.1050438>.
14. Bröder J, Okan O, Bauer U, et al. Health literacy in childhood and youth: a systematic review of definitions and models. *BioMed Central Public Health*. 2017;17 (1):1–25. <https://doi.org/10.1186/s12889-017-4267-y>.
15. Sarhan MB, Fujiya R, Kiriya J, et al. Health literacy among adolescents and young adults in the eastern mediterranean region: a scoping review. *British Medical Journal Open*. 2023;13(6), e072787. <https://doi.org/10.1136/bmjopen-2023-072787>.
16. Ayaz-Alkaya S, Kulakçı-Altuntaş H. Nutrition-exercise behaviors, health literacy level, and related factors in adolescents in Turkey. *J Sch Health*. 2021;91(8): 625–631. <https://doi.org/10.1111/josh.13057>.
17. Caldwell EP, Melton K. Health literacy of adolescents. *J Pediatr Nurs*. 2020;55: 116–119. <https://doi.org/10.1016/j.pedn.2020.08.020>.
18. Jafari A, Sany SBT, Peyman N. The status of health literacy in students aged 6 to 18 old years: a systematic review study. *Iran J Public Health*. 2021;50(3):448. <https://doi.org/10.18502/ijph.v50i3.5584>.